Lab 3.4-5

Question 1

I would construct the matrix based on the camera position, target and up vector

Question 2

Shadow mapping uses depth comparisons (z) to determine whether a fragment is in shadow. Environment mapping uses texture lookups based on direction vectors and does not involve depth comparisons.

Question 3

Because it does not account for the depth discontinuities that can happen at shadow edges.

Question 4

First person games like shooters, because of the complex and dynamic geometry that still requires accurate shadows.

Question 5

- Displacement Mapping: Better for close objects where the geometry needs to be very accurate like architectural views
- Normal Mapping: Better for distant objects and better performance like in shooters.

Question 6

It would contain the actual 3D normal vectors relative to the objects coordinate system. The mao would not just be blue instead if would have a bigger color distribution based on its position.